AP CALCULUS AB

Unit 1 • Worksheet 1-9A

# Worksheet 1-9A • IVT, EVT, and MVT

Name: _____ Date: _____

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

### Question 1

NO CALCULATOR

A runner's distance $D(t)$ is differentiable. "Was the runner ever going exactly as fast as her average speed?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) None

### Question 2

NO CALCULATOR

The height $h(t)$ of a kite is continuous on $[0, 30]$. "Was the kite ever exactly 25 m high?" given $h(0) = 2$ and $h(30) = 40$. Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) None

### Question 3

NO CALCULATOR

"Was there a highest point of the kite on $[0, 30]$?" Which theorem answers this?

(A) IVT
(B) MVT
(C) None
(D) EVT

### Question 4

NO CALCULATOR

Which theorem needs a **closed** interval and gives an absolute maximum and minimum?

(A) IVT
(B) EVT
(C) MVT
(D) All three

**Questions 5 to 7 refer to the table. $f$ is differentiable.**

| $x$ | 0 | 3 | 5 | 8 |
|---|---|---|---|---|
| $f(x)$ | 2 | $-4$ | 6 | 6 |

### Question 5

NO CALCULATOR

Which statement is guaranteed?

(A) $f(c) = 0$ for some $c$ in $(5, 8)$
(B) $f(c) = 0$ for some $c$ in $(0, 3)$
(C) $f'(c) = 0$ for some $c$ in $(0, 3)$
(D) $f'(c) = -2$ for some $c$ in $(3, 5)$

### Question 6

NO CALCULATOR

The MVT guarantees a $c$ in $(3, 5)$ with $f'(c) =$

(A) $1$
(B) $-5$
(C) $5$
(D) $10$

### Question 7

NO CALCULATOR

Must there be a $c$ in $(5, 8)$ with $f'(c) = 0$?

(A) Yes, by the MVT.
(B) Yes, by the IVT.
(C) No, $f$ is constant there.
(D) No, the MVT needs $f(a) \neq f(b)$.

### Question 8

NO CALCULATOR

$h$ is continuous on $[0, 4]$ with a sharp corner at $x = 2$. Which theorems apply on $[0, 4]$?

(A) All three
(B) MVT only
(C) IVT and EVT only
(D) None

**Question 9** (Free response) NO CALCULATOR

$g$ is differentiable.

| $x$ | 1 | 2 | 4 | 7 |
|---|---|---|---|---|
| $g(x)$ | 6 | 1 | 5 | 11 |

(a) Must $g(c) = 3$ for some $c$ in $(1, 2)$? Justify.
(b) Must $g'(c) = 2$ for some $c$ in $(2, 4)$? Justify.
(c) Must $g$ have an absolute minimum on $[1, 7]$? Justify.
(d) $k(x) = \frac{1}{x-3}$ has $k(2) = -1$ and $k(4) = 1$. Does the IVT guarantee $k(c) = 0$ in $(2, 4)$? Explain.